



INDIVIDUAL DESIGN JOURNAL – LUNAR ROVER WHEELS

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Team members

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Problem Statement

The design and justify the creation of a set of wheels that should be lightweight and durable for a six-wheel lunar rover. This rover should have the ability to traverse abrasive and loose lunar regolith. This wheel must fit within the constraints provided such as fitting within the mandated diameter range (80-130mm) and have the ability to integrate with the specified 12 V drive motor/geartrain. It should allow the rover to meet speed band requirements of 0.28-0.37 km/h through the correct selection of wheel diameter and coordination with gearing.

Background

Due to the lunar rover operating on the moon there will be unique engineering challenges such as there being low gravity (1.62m/s^2) which therefore decreases traction and normal force. Must also be aware of the lunar regolith as it is sharp, fine and very abrasive therefore requiring the materials to be able to resist wear. (Gaier, 2005, NASA Technical Memorandum). “Lunar dust, easily perturbed and dispersed, adheres and abrades materials due to its rough and irregular grain morphology.” (NASA, 2025) On the moon there is also a vacuum and extreme temperatures; with

temperatures being between -173 °C to +127 °C. Due to this rubber tyres would have to be ruled out as they would either crack, degrade or be outgassed. To prevent this from happening the Apollo rovers used woven steel mesh wheels. As seen in the NASA Apollo Lunar Roving Vehicle Wheel Technical report from 1971 most wheels are lightweight with rigid metallic wheels with machined tread patterns as they are standard for rover designs. (Johnson, S. W., & Chua, K. M. (1993))

Objectives and Functional Requirements

Objectives	Functional
Lightweight: to ensure that the wheel mass is minimised to keep rover under 20kg	Diameter: needs to be within 80-130mm (selected 130mm)
High durability: to ensure rover can withstand repeated loading cycles and abrasive lunar regolith	Load capacity: each wheel must support at least 5.4N
Traction and mobility: have good sufficient grip to make sure there are no wheel slips	
Speed: ensure rover can travel within the speed band of 0.28-0.37km/h	

Constraints

The wheel had to stay within the requirements of 80-130mm. To ensure that the speed requirement could be met (add speed requirement) a 130mm wheel was chosen. The following calculations demonstrate why this is necessary:

Assumptions:

- D = 3cm

- 220g/ per trip
- $V_{min} = 0.28\text{km/h} = 280\text{m/hr}$
- $V_{max} = 0.37\text{ km/h} = 370\text{m/h}$
- $D_{min} = 80\text{mm}$
- $D_{max} = 130\text{mm}$

$$Volume = \frac{4}{3}\pi r^3 \left(\frac{3}{2}\right)^3$$

$$Volume = 14.13\text{cm}^3$$

$$\rho = 1.5\text{g/cm}^3$$

$$m^1 = P v = 1.5 \times 14.13$$

$$m^1 = 21.2g$$

$$N^1 = \frac{m_{total}}{m^1} = \frac{200}{21.2} = 9.43 \approx 1$$

$$S^1 = 300m$$

$$S^2 = 2 \times 300 = 600m$$

$$U = 280m/h$$

$$v = \frac{s^1}{t} = \frac{\pi d \times RPM \times 60}{1}$$

$$d = \frac{V}{\pi \times RPM \times 60} = \frac{280}{\pi \times 10 \times 60}$$

$$d = 0.1486m = 148.6mm$$

$$V = \frac{s^1}{t} = \frac{\pi d \times RPM \times 60}{t} = \frac{\pi \times 130 \times 10 \times 60}{1}$$

$$v = \frac{244.9m}{hr}$$

$$v = \frac{s}{t} \Rightarrow t = \frac{s}{v} = \frac{600}{280} = 2.14hr$$

$$v^1 = \frac{280m}{hr} \times 2t \times 2.14hr = 11$$

$$v^2 = 444, t = \frac{s}{v} = \frac{600}{280} = 2.45hr$$

$$v^2 = \frac{244m}{hr}, t = 2.45hr = 12.5$$

$$t = 12.5 \times S \times F$$

$$t = 12.5 \times 1.5$$

Due to 130mm being the biggest the diameter can be to reduce the diameter the gearbox speedup is necessary to meet requirement. With the speed requirements being a speed band of 0.28 – 0.37km/h

$$C = \pi D$$

$$C = \pi \times 0.130 \approx 0.408m$$

$$Speed(v) = RPM \times C \times \frac{60}{100}$$

$$v = 10 \times 0.408 \times \frac{60}{1000} \approx 0.245 km/h$$

$$0.28km \text{ RPM} = \frac{0.28}{0.02448} = 11.44RPM$$

Meaning a 1.14 gear up

$$0.37km \text{ RPM} = \frac{0.37}{0.02448} \approx 15.11RPM$$

Meaning a 1.51 gear up

Brainstorming

Sketch:

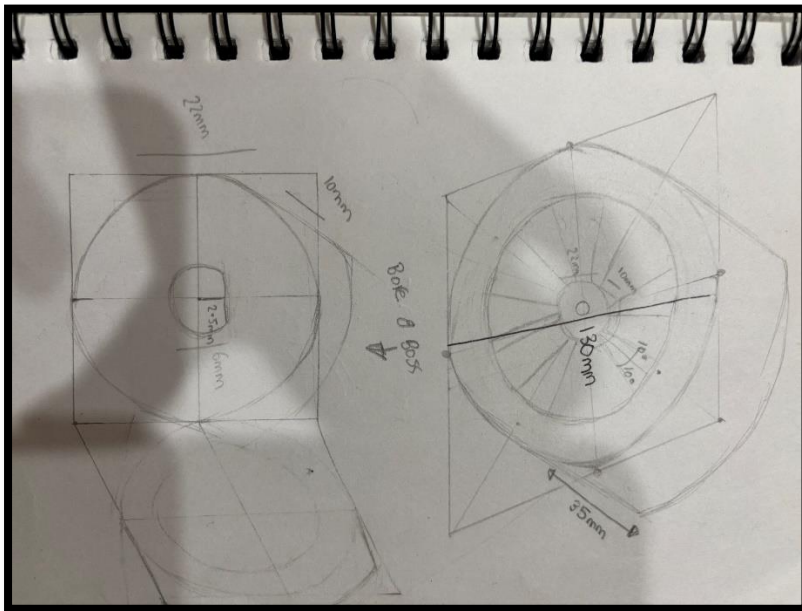


Figure 1: initial sketch

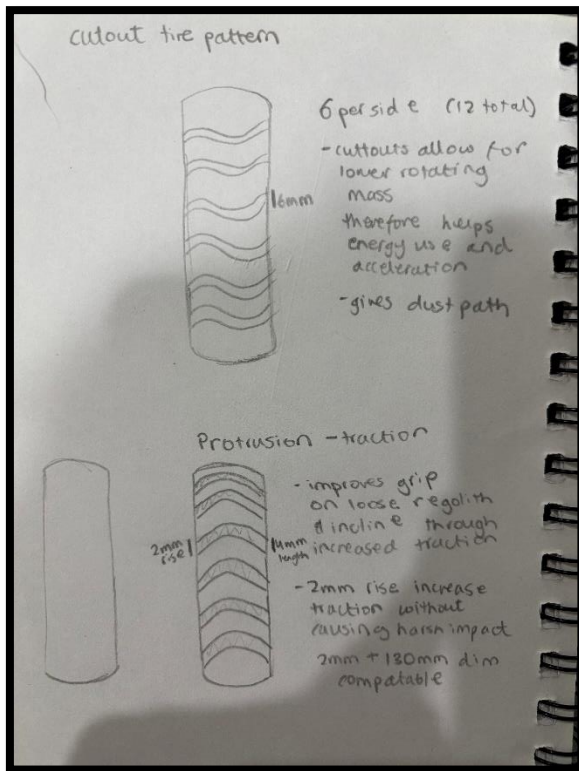


Figure 2: initial tire pattern sketch

CAD:

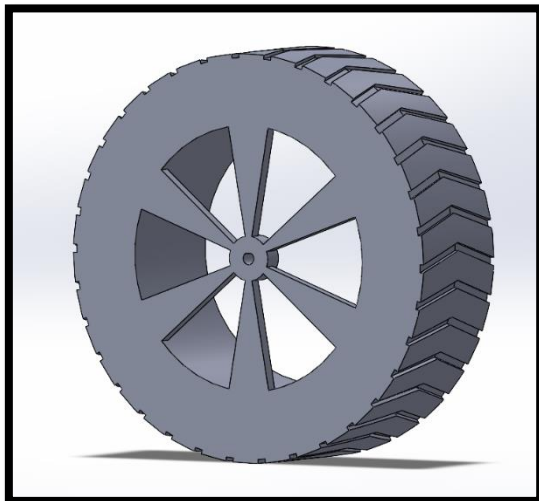


Figure 1: chevron/Arrow Tread Pattern

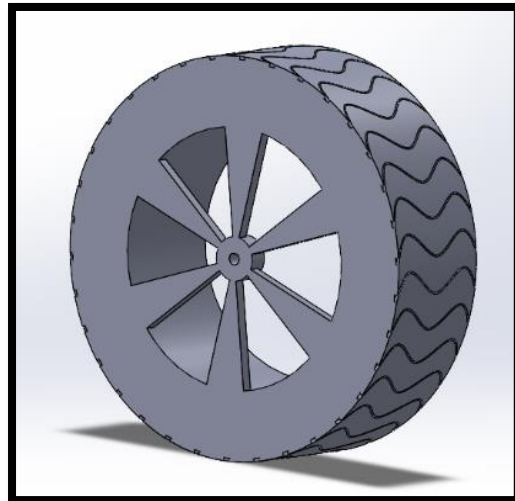


Figure 2: Wavy/Sine Tread Pattern

Pros and cons for Figure 1

Pros

Cons

Is very good and self-cleaning the loose regolith because of the angled grooves pushing dust sideways	It is slightly more complex geometry to machine
Creates a forward type of padding effect that help to improve traction when on slopes	Potentially could wear unevenly if not supported by a stiff rim
Makes an even load transfer across the contact patch	
Is proven in rover wheels and off-road vehicles for traction	

Pros and cons for Figure 2

Pros	Cons
Is a simplified cut path and therefore easier to manufacture/create in CAD	It is less efficient in channelling dust away and therefore can cause the lunar regolith to create clogging
It creates multi directional traction meaning it is equal in forward and lateral directions	It would provide a weaker directional traction compared to figure 1
It helps reduce the stress concentration because curves avoid sharp corners	

Final choice – Figure 1:

For the final choice figure 1 was selected due to the rover having to operate on loose and abrasive lunar regolith where both traction and self-cleaning is incredibly important to ensure/avoid wheel slips and any loss of mobility. The design of figure 1 allows dust to be channelled away which therefore provides superior forward grip which has been proven through the NASA Apollo rover wheels, as well as the modern Mars rover prototypes. Although Figure 2 would be easy to manufacture, figure 1 is better suited to meet the functional requirements of traction, durability and efficient load transfer. (Asnani, V., Delap, D., & Creager, C. (2009)). Despite this both wheels have their pros and cons with as said in a NASA report by Clark P.E “that certain test variables were more significant than others in determining performance for each type of test, and no single wheel design clearly outperformed the others in all areas.”

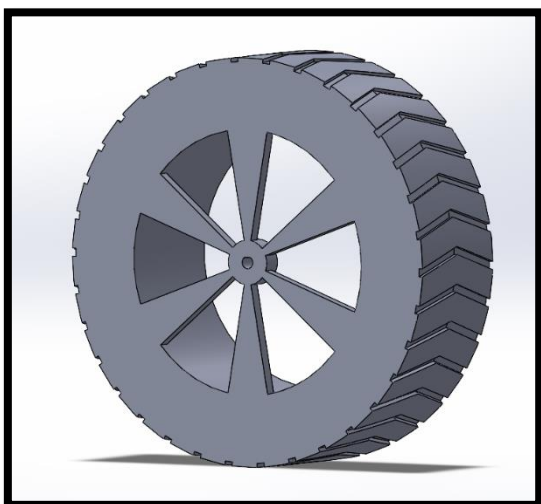
Material selection:

The following table was used to compare the alternative options for materials.

Materials	Density (kg/m ³)	Mass per whell	Pros	Cons
Aluminium	2700	116g	Strong Light Easy to apply in CAD Abrasion resistant when anodised	Is slightly weaker than titanium
Steel	7850	343g	Strong Cheap	Would be too heavy
Titanium	4430	194g	Strong Corrosion resistant	Can be expensive (despite this project having no cost requirement) Heavier
Carbo fibre	~1600	70g	Lightest of all options Strong	Would be harder to justify CAD Risk of delamination

Due to the following seen above Aluminium 6061-T6 was chosen as it balances weight, has good durability, has easy manufacturability and is CAD verified. This material is also mass efficient being 116g per wheel. It has also been proven to work in other aerospace applications, with its hard anodising providing good abrasion resistance against the lunar dust.

CAD Model

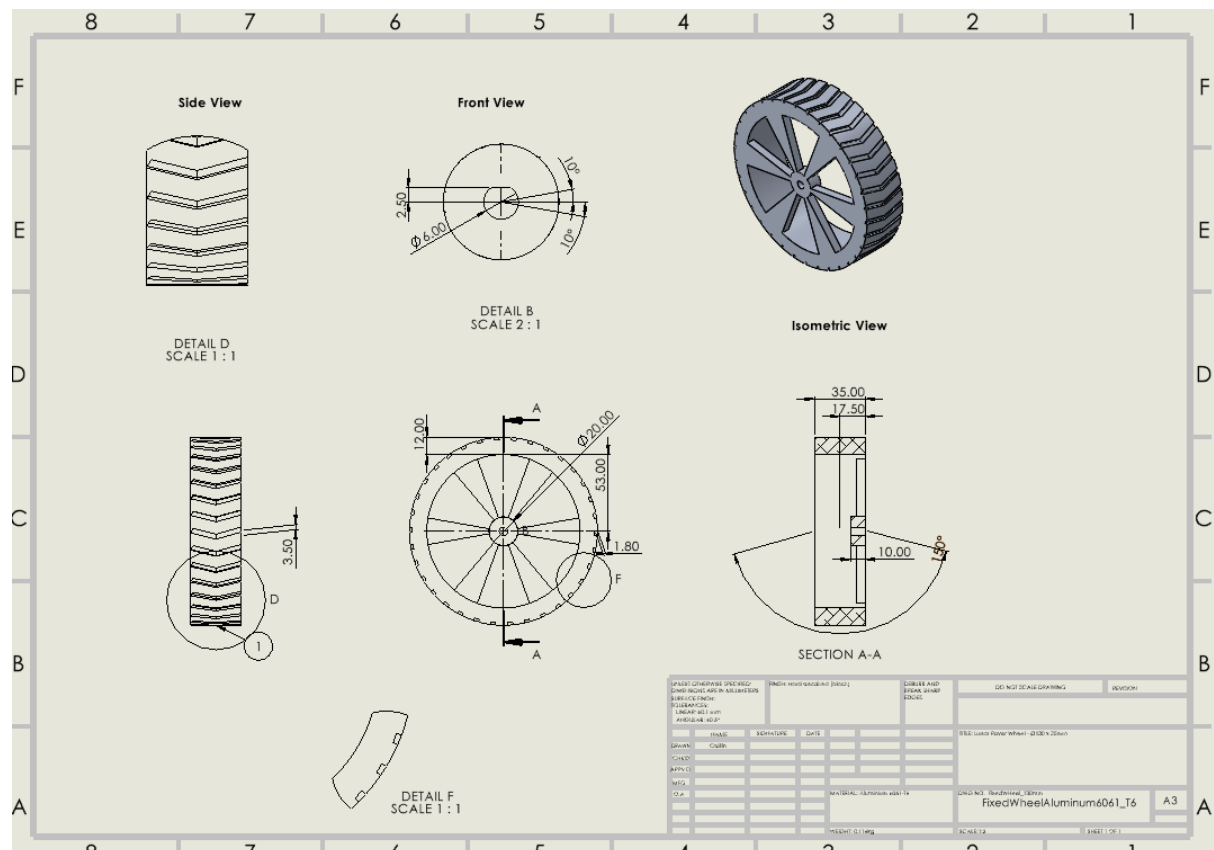


Finalised Wheel dimensions

- Wheel: 130mm x 35mm
- Boss: 22mm x 10 mm thick
- Bore: 6mm (placeholder/ can be changed if needed for motor shaft)
- Spokes: x6 spokes, 10 degree sweep, 3mm thick x 10mm wide

Wheel Technical Sketch

The following technical drawing demonstrates an orthographic view, section views, detail views and an isometric view.

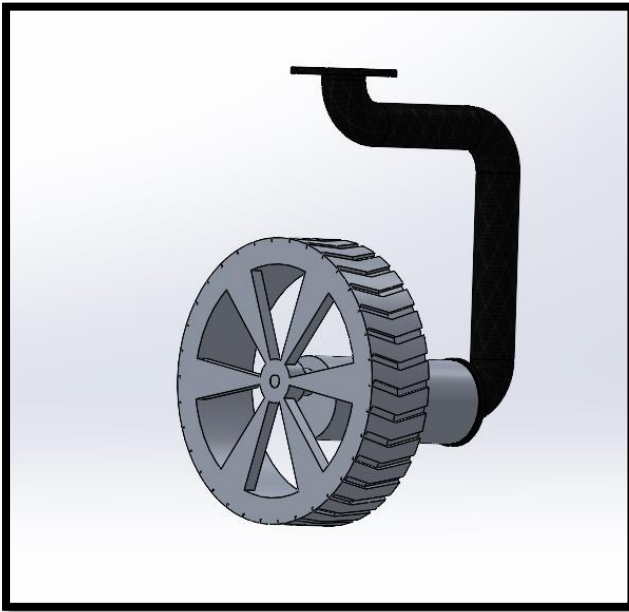


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR 0.1mm ANGULAR 0.5		FINISH: Machined, Hard-Anodised		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION	
DRAWN: Caitlin Frank		SIGNATURE: caitlin		DATE: 11/09/25		TITLE: Lunar Rover Wheel			
CHK'D:									
APP'D:									
MFG: RMIT Group 18									
Q.A:									
				MATERIAL: Aluminium 6061-T6		DWG NO. Group18-Caitlin-Wheel-DetailedDesignModel		A3	
				WEIGHT: ~0.116kg		SCALE: 1:2		SHEET 1 OF 1	

Wheel and Drive system sub assembly Technical Sketch

The sub assembly has combined both the wheel and drive motor and gearbox housing and creates the completed drive module. This design makes sure that the wheel is able to be powered independently while also maintain durability and manufacturability.

The sub assembly was made to have lightweight construction and have structural rigidity. Through the motor housing and wheel hub being integrated into a compact design the system therefore makes sure the design is simplified and stops any unnecessary bulk.



Wheel kinematics and speed – to meet speed requirement of 0.28 – 0.37

References

Asnani, V., Delap, D., & Creager, C. (2009). The Development of Wheels for the Lunar Roving Vehicle. *Ntrs.nasa.gov*. <https://ntrs.nasa.gov/citations/20100000019>

Johnson, S. W., & Chua, K. M. (1993). Properties and Mechanics of the Lunar Regolith. *Applied Mechanics Reviews*, 46(6), 285–300.
<https://doi.org/10.1115/1.3120358>

NASA. (2025). *Abrasive effects of lunar regolith on material wear for long-term lunar applications* (NASA/TM–20250000687). NASA Technical Reports Server.
<https://ntrs.nasa.gov/citations/20250000687>

Clark, P. E., & Zacny, K. (2024). *ISRU pilot excavator wheel testing in lunar regolith simulant* (NASA/TM–20240001016). NASA Technical Reports Server.
<https://ntrs.nasa.gov/citations/20240001016>

Meeting Minutes:

Meeting - 14/08/2025

Week Number: 4	
Meeting Topic	Rover Component Assignment
Place	EJ's cafe – Building 56
Attendees	

Student ID	Name	Attendance
4006866	Claire Schubert	Yes
4090482	Esha Macherla	Yes
4097017	Shreya Mahendiratta	Yes
4091918	Caitlin Frank	Yes
4137391	Sugreev Aravindhan	Yes
4190482	Oliver Wilde	Yes

Agenda Details
Drive system - Esha Body - Sugreev Wheel - Caitlin Powerpack & Solar Panel - Oliver Suspension - Claire Camera Housing - Shreya

Decisions			
Action Plan	Responsibility	Deadline	Status
Complete problem statement (3.2) and brainstorming (3.3) for assigned component	All	21/08	

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Meeting – 21/04/2025

Week Number: 5		
Meeting Topic		Rover Component Assignment
Place		Teams meeting
Attendees		
Student ID	Name	Attendance
4006866	Claire Schubert	Yes
4090482	Esha Macherla	Yes
4097017	Shreya Mahendiratta	Yes
4091918	Caitlin Frank	Yes
4137391	Sugreev Aravindhan	N/A
4190482	Oliver Wilde	Yes

Agenda Details
Drive system - Esha Body - Sugreev Wheel - Caitlin Powerpack & Solar Panel - Oliver Suspension - Claire Camera Housing – Shreya Discussing each other's parts and going through our sketches and CAD designs.

Decisions			
Action Plan	Responsibility	Deadline	Status
Complete problem statement (3.2) and brainstorming (3.3) for assigned component	All	21/08	Completed
Complete individual preliminary designs as a sketch	All	24/08	Completed
Complete CAD preliminary design	All	1/09	

Meeting – 28/08/2025

Week Number: 6		
Meeting Topic		Rover Component Attachments
Place		Teams meeting/B56 Cafeteria
Attendees		
Student ID	Name	Attendance
4006866	Claire Schubert	Yes
4090482	Esha Macherla	Yes
4097017	Shreya Mahendiratta	Yes
4091918	Caitlin Frank	Yes
4137391	Sugreev Aravindhan	Yes
4190482	Oliver Wilde	Yes

Agenda Details
Drive system - Esha

Body - Sugreev
Wheel - Caitlin
Powerpack & Solar Panel - Oliver
Suspension - Claire
Camera Housing – Shreya
Discussing how components assemble together. Deciding dimensions of connecting components parts and weight considerations. Putting forward questions and concerns regarding overall rover constraints.

Decisions			
Action Plan	Responsibility	Deadline	Status (overall)
Complete problem statement (3.2) and brainstorming (3.3) for assigned component	All	21/08	Completed
Complete individual preliminary designs as a sketch	All	24/08	Completed
Complete CAD preliminary design	All	1/09	Pending

Status (Yes/No)

Action Plan	Due	Caitlin	Oliver	Shreya	Claire	Esha	Sugreev
Complete problem statement (3.2) and brainstorming (3.3) for assigned component	21/08	Yes	Yes	Yes	Yes		Yes
Complete individual preliminary designs as a sketch	24/08	Yes	Yes	Yes	Yes		Yes
Complete CAD preliminary design	1/09		Yes	Yes	Yes		Yes

Meeting – 1/09/2025

Week Number: 6 (Second meeting week 6)		
Meeting Topic		CAD Model (Preliminary)
Place		Home (Teams Meeting)
Attendees		
Student ID	Name	Attendance
4006866	Claire Schubert	Yes
4090482	Esha Macherla	Yes
4097017	Shreya Mahendiratta	N/A

4091918	Caitlin Frank	Yes
4137391	Sugreev Aravindhan	Yes
4190482	Oliver Wilde	Yes

Agenda Details	
Drive system - Esha Body - Sugreev Wheel - Caitlin Powerpack & Solar Panel - Oliver Suspension - Claire Camera Housing – Shreya Share CAD Models. Attempt Sub-assembly to practice. Next Milestone: Complete Technical Drawing for CAD models.	

Decisions			
Action Plan	Responsibility	Deadline	Status (overall)
Complete problem statement (3.2) and brainstorming (3.3) for assigned component	All	21/08	Completed
Complete individual preliminary designs as a sketch	All	24/08	Completed
Complete CAD preliminary design	All	1/09	Completed
Complete Technical Drawing	All	14/09	Pending

Status (Yes/No)							
Action Plan	Due	Caitlin	Oliver	Shreya	Claire	Esha	Sugreev
Complete problem statement (3.2) and brainstorming (3.3) for assigned component	21/08	Yes	Yes	Yes	Yes	Yes	Yes
Complete individual preliminary designs as a sketch	24/08	Yes	Yes	Yes	Yes	Yes	Yes
Complete CAD preliminary design	1/09	Yes	Yes	Yes	Yes	Yes	N/A

Meeting – 11/09/2025

Week Number: 7	
Meeting Topic	Rover-sub assembly (addressing any major issues)
Place	B56 Cafeteria (Teams Meeting)
Attendees	

Student ID	Name	Attendance
4006866	Claire Schubert	Yes
4090482	Esha Macherla	Yes
4097017	Shreya Mahendiratta	Yes
4091918	Caitlin Frank	Yes (online)
4137391	Sugreev Aravindhan	N/A
4190482	Oliver Wilde	Yes

Agenda Details
Found issues with sub-assembly height. Re-attempting adjustments to rover sub-assembly and working on technical drawings to get accurate dimensions. Discussed team collaboration, concerns regarding dimensions and design constraints and non-communicating group members. Next meeting: next Sunday 14/09 (flexible in –person or online). Next Milestone: Complete Technical Drawing for CAD models. Start working on sub-assembly to check for any issues meeting design requirements

Decisions			
Action Plan	Responsibility	Deadline	Status (overall)
Complete problem statement (3.2) and brainstorming (3.3) for assigned component	All	21/08	Completed

Complete individual preliminary designs as a sketch	All	24/08	Completed
Complete CAD preliminary design	All	1/09	Completed
Complete Technical Drawing	All	14/09	Pending
Rover Sub-assembly	All	14/09	Pending

Status (Yes/No)							
Action Plan	Due	Caitlin	Oliver	Shreya	Claire	Esha	Sugreev
Complete problem statement (3.2) and brainstorming (3.3) for assigned component	21/08	Yes	Yes	Yes	Yes	Yes	Yes
Complete individual preliminary designs as a sketch	24/08	Yes	Yes	Yes	Yes	Yes	Yes
Complete CAD preliminary design	1/09	Yes	Yes	Yes	Yes	Yes	Yes (Incomplete)
Complete Technical Drawings	14/09						
Complete Rover-sub assembly	14/09						

Meeting – 14/09/2025

Week Number: 7 (Second Meeting week 7)		
Meeting Topic		Rover-sub assembly (addressing any major issues)
Place		Teams Meeting
Attendees		
Student ID	Name	Attendance
4006866	Claire Schubert	Yes
4090482	Esha Macherla	Yes
4097017	Shreya Mahendiratta	Yes
4091918	Caitlin Frank	Yes
4137391	Sugreev Aravindhan	Yes
4190482	Oliver Wilde	Yes

Agenda Details
Found issues with sub-assembly height. Fixed and refined dimensions. Shared all CAD models. Some technical drawings were shared, due 16/09. Adding screws and fastner holes for the body is an exception today.
Next meeting: next Tuesday 16/09 (online).
Next Milestone: Complete Technical Drawing for CAD models.

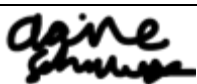
Decisions			
Action Plan	Responsibility	Deadline	Status (overall)

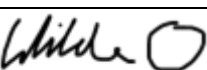
Complete problem statement (3.2) and brainstorming (3.3) for assigned component	All	21/08	Completed
Complete individual preliminary designs as a sketch	All	24/08	Completed
Complete CAD preliminary design	All	1/09	Completed
Complete Technical Drawing	All	16/09	Pending
Rover Sub-assembly	All	14/09	Completed

Status (Yes/No)							
Action Plan	Due	Caitlin	Oliver	Shreya	Claire	Esha	Sugreev
Complete problem statement (3.2) and brainstorming (3.3) for assigned component	21/08	Yes	Yes	Yes	Yes	Yes	Yes
Complete individual preliminary designs as a sketch	24/08	Yes	Yes	Yes	Yes	Yes	Yes
Complete CAD preliminary design	1/09	Yes	Yes	Yes	Yes	Yes	Yes (missed deadline)

Complete Technical Drawings	16/09						
Complete Rover-sub assembly	14/09	Yes	Yes	Yes	Yes	Yes	Missed deadline (future version uploaded)

Contribution Table

Group 18					
Name	Student No	Student Signature	Contribution		
			Part-A Preliminary Model Deadline	Part-A Detailed Model Deadline	No of Meetings Attended (Wk 3 to 8)
Claire Schubert	4006866		Not met, however mostly completed – just not finalized due to body not being completed on time	Met	6
Esha Macherla	4090482		Met	Met	6

					
Caitlin Frank	4091918		Met	Met	6
Shreya Mahendiratta	4097017	श्रेया	Not met, however mostly completed – just not finalized due to body not being completed on time	Met	5
Oliver Wilde	4190482		Met	Met	6
Sugreev Aravindhana	 4137391		Not met	Met	4